Incomplete detector calibration

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2. Independent result: exact detector partial-identification interval

Let trusted density matrices rho_i (i=1,...,m) have exact measured response rates p_i for an otherwise unknown but stable binary measurement effect 0 <= E <= I on a finite-dimensional Hilbert space. For a target state rho_T define

```
L(p) = min Tr(E rho_T),   U(p) = max Tr(E rho_T),
subject to Tr(E rho_i)=p_i for every i and 0 <= E <= I.
```

These are finite semidefinite programs, and by definition are the sharp possible target-response bounds for the entire admissible effect class. Both extrema are attained when the calibration vector is feasible because the effect set is compact. The image is a closed interval by convexity. **They are data dependent**: a target outside the real reference span does not imply ambiguity for every possible calibration dataset.

For any real vector lambda, put A=lambda-weighted subtraction `A=rho_T-sum_i lambda_i rho_i`. Then

```
Tr(E rho_T) = sum_i lambda_i p_i + Tr(E A)
            <= sum_i lambda_i p_i + Tr(A_+),
```

where `A_+` is the positive part of A and the inequality is sharp when E is its positive spectral projector if no other constraints are imposed. Consequently,

```
U(p) <= inf_lambda [sum_i lambda_i p_i + Tr((rho_T-sum_i lambda_i rho_i)_+)].
```

The inequality is equality under the usual finite-dimensional Slater condition: at least one compatible effect satisfies 0 < E < I. At boundary response vectors, rely on the primal SDP or limiting dual, not an unqualified interior-strong-duality assertion. For normalized densities, `L(p)=1-U(1-p)` follows by exchanging E and I-E.

Elementary complete proof for a qubit

Use two trusted reference states |0><0| and |1><1|. Suppose their calibrated click probabilities are p0,p1. A possible effect has the form

```
E = [[p0, z], [conjugate(z), p1]].
```

Positivity of E and I-E is equivalent to

```
|z| <= R := min(sqrt(p0*p1), sqrt((1-p0)*(1-p1))).
```

For the held-out target |+>=(|0>+|1>)/sqrt(2), `q=(p0+p1)/2+Re(z)`. Choosing z=+/-R proves the *exact* interval

```
(p0+p1)/2-R <= q <= (p0+p1)/2+R.
```

In particular, if p0=p1=p, the range is `[max(0,2p-1),min(1,2p)]`. At p=0 the calibration forces E=0 and q=0, despite |+><+| lying outside the real reference span. At p=1/2 the same two reference probabilities are compatible with every target rate from zero to one. These are extreme physical effects, not a numerical fitting artifact.

This is an instance of established quantum partial tomography and set-identification SDPs (e.g. *Physical Review A* 98, 042318, 2018), not a claim of mathematical originality. Its application to the present measurement contract is useful because it avoids inferring universal ignorance from mere failure of reference informational completeness.

Uncertain reference data

When independently justified intervals [l_i,u_i] replace exact p_i, use the same primal with inequalities. Its Lagrangian bound reads

```
U <= inf_{beta_i,gamma_i>=0} [sum_i (beta_i u_i-gamma_i l_i)
      + Tr((rho_T - sum_i (beta_i-gamma_i)rho_i)_+)].
```

Equality follows under appropriate strict feasibility. This is how uncertainty and physicality should jointly enter, rather than treating a noisy point estimate as exact.

3. Cross-check: tetrahedral source-response bound

For four trusted tetrahedral qubit states rho_i=(I+r_i dot sigma)/2, define D_i=(3rho_i-I)/2. They form a dual frame and satisfy sum D_i^2=(5/2)I. The sixteen probabilities p_ij=Tr(E rho_i tensor rho_j) determine E exactly. Under entrywise errors |delta p_ij| <= delta, the *maximum response over separable states*, b_sep(E), changes by at most 4 delta, which is sharp. The proof uses the dual weights

```
w_i(x) = Tr(D_i rho_x)=(1+3r_i dot x)/4.
```

The exact maximum of sum_i |w_i(x)| is two for Bloch vectors |x|<=1, attained at a tetrahedral antipode, with weights (-1/2,1/2,1/2,1/2). Product response weights therefore have absolute sum at most four. Direct construction of a sign-perturbation near E=I/2 attains the bound without leaving the effect set for sufficiently small delta.

Using alpha=.05, Delta=13/1280, and seventeen independent settings of n trials each, the sufficient requirement n > 50 log(34/alpha)/Delta^2 gives n=3,161,479, total 53,745,143. This is a *conservative sufficient* shot budget with trusted preparations, stable effect, and independent Bernoulli trials, not a necessary hardware cost.